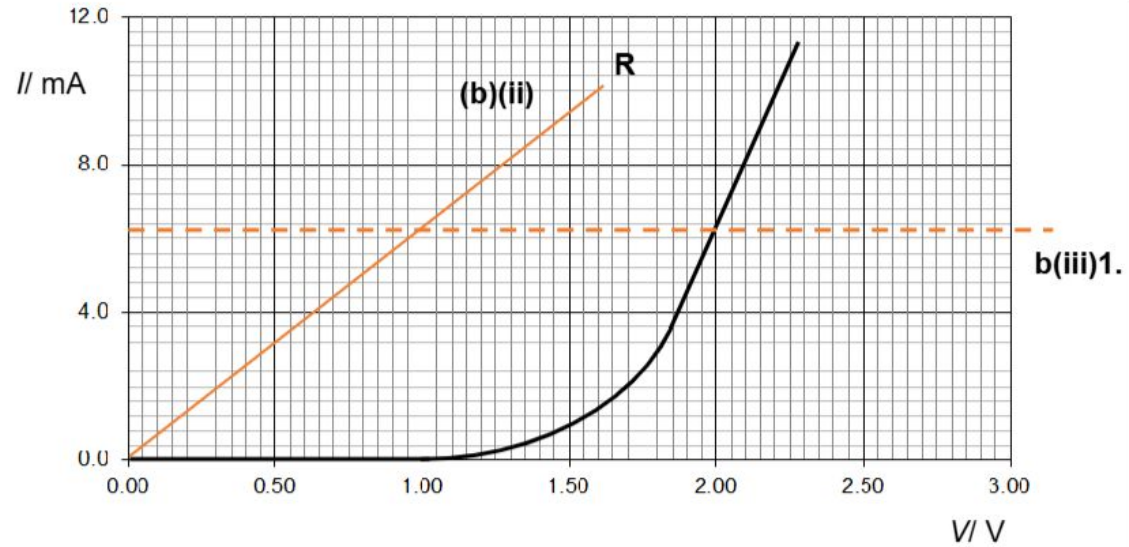
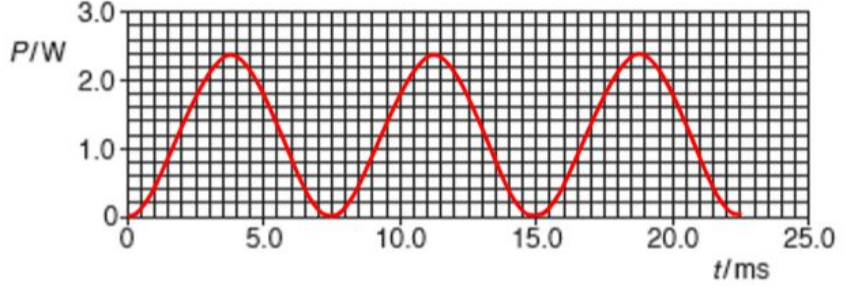
	The ball will rebound with a smaller (horizontal) speed.	B1
9a(i)	$R = \frac{\rho l}{A}$ $= \frac{5.0\times10^{-7}\times2.0}{3.3\times10^{-7}}$ $= 3.03\ \Omega$ $= 3.0\ \Omega$	 A1
9aii	*Current through wire* = $\frac{1.5}{3.0+0.75}$ $= 0.40$ A $I = nAqv$ v = $\frac{0.40}{8.5\times10^{28}\times3.3\times10^{-7}\times1.6\times10^{-19}}$ $= 8.91\times10^{-5}$ m s^{-1}	C1 A1
9aiii	When the length of the wire is doubled, the <u>resistance</u> $R(=\frac{\rho l}{A})$ <u>of the second wire is twice</u> that of the first wire. Since $I = \frac{E}{R+r}$, <u>current through the second wire decreased</u>. Using $v = \frac{I}{nAq}$, Since the number density (as the material and temperature are the same), cross-sectional area and the charge of charge carrier (electron) are unchanged, <u>the average drift velocity of the electrons decreased.</u>	M1 M1 A1
9bi	*Using* $R = \frac{V}{I}$ = 2.25/0.0108 = 208 Ω	 M1 A1

9bii	 Line R should be a straight line passing through (0,0) and (1.60 V, 10.0 mA)	B1
9biii 1.	Draw a horizontal line on Figure 9.2 such that the pd across 160 Ω and LED1 adds up to 3.0 V. I = 6.2 mA	B1 A1
9biii 2.	With two LEDs, effective resistance of the circuit is lowered, leading to higher current from battery.With higher current, the potential difference across the 160Ω resistor is larger, leaving less potential difference across the LEDs. By $P = \frac{V^2}{R_{LED}}$,the power of the LED will be lower OR With two LEDs in parallel, effective resistance between the LEDs is lowered. By Potential Divider Principle, the potential difference across each LED is also lowered. By $P = \frac{V^2}{R_{LED}}$,the power of the LED will be lower OR From the graph, it can be seen the current (of the LED) decreases with potential difference, so the power (P=VI) dissipated in each LED is lower, so brightness is dimmer.	M1 M1 A1 (M1) (M1) (A1)
9ci	V_s / V_p= N_s / N_p N_s = 52/150 x 1200 = 416 turns	M1 A1
9cii	$V_{rms} = \frac{V_0}{\sqrt{2}} = \frac{52}{\sqrt{2}}$ = 36.8 V mean power = $\frac{V_{rms}^2}{R} = \frac{(36.8)^2}{R}$ $R = \frac{(36.8)^2}{1.2}$ = 1100 Ω	C1 A1

9ciii	sine-squared graph with troughs at zero power and at least 3 'cycles' peak power = 2.4 W and correct period 	B1 B1